

Edge cases (empty vs. missing) — Hints

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Only 2 hints — work through them in order, and don't jump ahead until you've genuinely tried. Each hint has 3 depth levels: **Basic** (the plain idea), **Intermediate** (proper Python), **Advanced** (how a real codebase would actually write it). Read Basic first even if you already know Python — it's the fastest way to spot exactly what each deeper level adds.

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Hint 1 — The idea, and try it yourself

Basic Version

There are two different kinds of "empty" here:

1. The `.env` file exists but has nothing written in it.
2. The `.env` file has `SOME_KEY=` — the key is there, but nothing comes after the `=`.

For each case, your config loader has to decide: should this count as "missing" (raise an error), or "empty but okay"? You get to decide, and should be able to explain why.

Try both cases yourself:

- Make a totally empty `.env` file (no text at all), run your loader, see what happens.
- Make a `.env` with `OPENAI_API_KEY=` (equals sign, nothing after it), run your loader, see what happens.

Check: does `os.getenv("OPENAI_API_KEY")` give you `None`, or does it give you `""` (an empty piece of text)? That difference is the key to this whole exercise.

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Intermediate Version

Python's `os.getenv()` already treats these two cases differently, on its own — spotting that is the whole exercise:

```
os.getenv("OPENAI_API_KEY")    # key never set at all      -> returns None
os.getenv("OPENAI_API_KEY")    # file has "OPENAI_API_KEY=" -> returns ""
```

`None` and `""` are two different values in Python. `None` means "this key was never set." `""` (an empty string) means "this key was set, but to nothing." Your `require_env()` has to decide whether *both* of those should raise `MissingConfigError`, or only the first.

Test both situations against your own `require_env()` (or equivalent) from the Build Task, and inspect the value precisely instead of guessing:

```
value = os.getenv("OPENAI_API_KEY")
print(repr(value))    # None -> prints: None
                    # "" -> prints: ''
print(type(value))    # <class 'NoneType'> or <class 'str'>
```

Using `repr()` (instead of a plain `print(value)`) shows you the quotes around a string, so you can't mistake an empty string `''` for anything else.

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Advanced Version

This is really a small version of a general problem: "missing" and "present but empty" are not the same thing, and collapsing them together (or keeping them apart) is a real design decision — not just a syntax detail. A careless check like `if not value:` treats `None`, `""`, and even things like `"0"` or a whitespace-only string `" "` all as equally "falsy," which quietly hides distinctions that matter. For a secret like an API key, you usually want to be explicit about exactly *which* falsy-looking values are invalid, rather than letting Python's truthiness rules make that call for you.

A production loader usually needs to handle a 3rd real case too: a value that's technically non-empty but still useless — like `OPENAI_API_KEY=` (only whitespace) or someone accidentally writing the literal text `"None"`. Testing your loader against a few of these messier inputs (and deciding whether `.strip()`-ing the value before checking it is the right call) is how a strict/lenient decision holds up outside the two textbook cases the exercise names.

Difference between Basic, Intermediate, and Advanced: Basic names the two situations in plain English and just says "try it and look." Intermediate shows exactly what `os.getenv()` returns for each one, in real code, and how to inspect it precisely with `repr()` and `type()` so you're certain what you're looking at. Advanced asks the harder question underneath — is a quick `if not value` check even safe, given how many different values Python treats as "falsy"? — and pushes past the two textbook cases into a few more realistic, messier inputs a loader might actually see in production.

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Hint 2 — The plan, and almost the whole thing

Basic Version

```
test 1: empty .env file
    run load_config()
    watch: does it raise an error, or does it keep going quietly?
    write down what happened, and whether that's what you want

test 2: .env has "OPENAI_API_KEY=" (nothing after the =)
    run load_config()
    watch: is the value None, or is it "" (empty text)?
    decide: should empty text count as "missing"? write down your answer and why
```

Treats empty text as "missing" (usually right for a secret):

```
value = os.getenv("OPENAI_API_KEY")
if value is None or value == "":
    raise MissingConfigError("...")
```

Treats empty text as "okay, just empty" (fine for some settings, not usually for a key):

```
value = os.getenv("OPENAI_API_KEY")
if value is None:
    raise MissingConfigError("...")
# empty text would pass through here without an error
```

Pick one for your loader and write one sentence saying why. Then check the [Solution](#).

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Intermediate Version

```
test 1: empty .env file
    run load_config()
    observe: does it raise MissingConfigError, or does it silently continue?
    write down which one happened, and whether that's what you want

test 2: .env contains "OPENAI_API_KEY=" (empty value)
    run load_config()
    observe: is the value None, or an empty string ""?
    decide: should an empty string count as "missing"? write down your answer and
```

If your current `require_env()` only checks `if value is None`, an empty string will slip through as "valid" — is that the behavior you actually want for a secret like an API key?

```
def require_env(key: str) -> str:
    value = os.getenv(key)
    if value is None or value == "":
        raise MissingConfigError(f"Required environment variable is missing: {key}")
    return value
```

What's missing: actual test calls, and the lenient version (which only checks `value is None`, with no `== ""` part). Write both yourself, and try them against an empty `.env` file and a `.env` with `OPENAI_API_KEY=`.

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Advanced Version

```
test 1: empty .env file          -> None -> raise, in both strict and lenient
test 2: "KEY="                   -> ""    -> decide: strict raises, lenient all
test 3: "KEY= " (only spaces)    -> "    " -> decide: does .strip() turn this in
test 4 (optional): instead of writing an if-check at all,
                        express the same rule declaratively -- e.g. a Pydantic
                        Field with min_length=1 -- so the validation library
                        enforces the rule instead of hand-written logic
```

```
from pydantic import BaseModel, Field

class Config(BaseModel):
    openai_api_key: str = Field(min_length=1)
```

This says the same rule as `if value is None or value == "":` — but as a constraint declared on the field itself, instead of an imperative statement someone has to remember to write correctly (and could accidentally write slightly differently in a 2nd place in the codebase).

Fill in the part that loads `os.getenv("OPENAI_API_KEY")` into this model and catches the validation error yourself, then compare all 3 of your finished versions against the [Solution](#).

Difference between Basic, Intermediate, and Advanced: same underlying idea (test the 2 textbook cases, then check the code) at 3 completeness levels — Basic and Intermediate test the 2 textbook cases the exercise names, as pseudocode and near-complete code respectively, with Intermediate adding the type contract real Python expects. Advanced adds a 3rd, more realistic case (whitespace-only), and replaces the `if` check entirely with a declared field constraint — the same rule, enforced by a validation library instead of hand-written logic, which is exactly what the Advanced solution builds.

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Full solution: [Show me the solution](#)